

Arishi Agarwal

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PROFESSIONAL SUMMARY

MSc Artificial Intelligence & Machine Learning candidate at Queen Mary University of London with hands-on experience building and deploying end-to-end ML systems: a **live RAG question-answering service** (FastAPI, FAISS, Docker, Hugging Face Spaces), **multi-modal deep learning classifiers** (ViT-B/16 + BioBERT), and production computer-vision pipelines from industry internships. Strong in **Python and PyTorch** with working MLOps practice (Docker, CI/CD, REST APIs). Seeking graduate Machine Learning / AI Engineer roles; eligible to work full-time in the UK.

EDUCATION

Queen Mary University of London
MSc Artificial Intelligence & Machine Learning
Symbiosis Institute of Technology, Pune
BTech Computer Science Engineering

Sept 2025 – Sept 2026
London, UK
2020 – 2024
Pune, India

RESEARCH & DISSERTATION PROJECT

Climate Risk Prediction System: Interpolation & Forecasting of Weather Data

2025 – 2026

Ongoing MSc Dissertation – Supervisor: Prof. Arthur Guillaumin

QMUL, London

- Designing an end-to-end **spatio-temporal forecasting system** that reconstructs missing weather data and quantifies predictive uncertainty over multivariate UK Met Office / NOAA series (temperature, humidity, wind speed, rainfall).
- Benchmarking **missing-data reconstruction** (linear interpolation, KNN imputation, **autoencoder** neural reconstruction) and multi-step (1–7 day) forecasts comparing **ARIMA/SARIMA** baselines against **LSTM** networks on MAE/RMSE, with robustness analysis under missing-data conditions.
- Integrating **uncertainty quantification** (Monte Carlo Dropout, prediction intervals) and delivering a reproducible, documented framework – reusable by future researchers – with dashboards that make forecasts and confidence intervals interpretable for non-technical stakeholders.

PUBLICATIONS

“**SneakerXchange: Enhancing User Experience through HCI Principles in a Reselling Website**” – co-author & presenter, *9th International Conference on Innovative Computing and Communication (ICICC-2026)*, University of Delhi with NIT Patna & University of Valladolid, Feb 2026.

TECHNICAL SKILLS

Languages: Python, SQL (MySQL, PostgreSQL), C++, UNIX/Linux Shell

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, CNNs, RNNs/LSTMs, Vision Transformers (ViT), Feature Engineering, Model Evaluation

LLMs & Generative AI: Hugging Face Transformers, BERT/BioBERT, Retrieval-Augmented Generation (RAG), FAISS & Vector Search, LangChain, OpenAI API, Prompt Engineering

MLOps & Deployment: Docker, REST APIs (FastAPI, Flask), Git, GitHub Actions (CI/CD), Hugging Face Spaces, AWS (S3, EC2)

Data & Visualisation: Pandas, NumPy, Time-Series Forecasting (ARIMA/SARIMA, LSTM), Statistical Analysis, EDA, Matplotlib, Seaborn, Tableau

SELECTED AI & DATA SCIENCE PROJECTS

FastAPI Docs RAG Assistant | *Python, FAISS, Streamlit, FastAPI, Docker, Hugging Face Spaces*

[Live Demo](#) | [GitHub](#)

- Built an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline over 153 FastAPI documentation files – ingesting, chunking (2,208 passages), embedding with **BGE-small-en-v1.5** and indexing with **FAISS** for sub-millisecond semantic search.
- Implemented dense, **hybrid BM25+dense (reciprocal rank fusion)** and **cross-encoder reranking** retrieval strategies; exposed a **FastAPI REST backend** (/ask, /health, /modes with Swagger UI) and a **Streamlit UI** with streaming answers and conversation history.
- Designed a reproducible **evaluation harness** (Recall@5, MRR, LLM-judged faithfulness/correctness); ablation study showed **BGE embeddings lifted Recall@5 0.75 → 0.92** and correctness **0.67 → 1.0**; **deployed live on Hugging Face Spaces** (Docker).

Multi-Modal Chest X-Ray Classifier | *PyTorch, ViT-B/16, BioBERT*

MSc Deep Learning Project

- Engineered an end-to-end multi-modal architecture fusing a **ViT-B/16** vision encoder with a **BioBERT** clinical-text encoder via **cross-attention** to classify 14 chest diseases from X-rays and radiology notes simultaneously.
- Built the full training pipeline – selective layer freezing, checkpointing and a forward pass returning 14-class logits and attention weights for interpretability.

Exoplanet Radius Prediction (1D CNN) | *TensorFlow, NASA Exoplanet Archive*

MSc AI/ML Coursework

- Trained a 3-block **1D CNN** on 3,847 confirmed exoplanet systems to predict planetary radius from 8 orbital/stellar features, achieving $R^2 = 0.921$ – beating MLP (0.887) and linear (0.743) baselines.
- Applied $\log_{10}$ transforms, StandardScaler and augmentation with EarlyStopping / ReduceLROnPlateau / ModelCheckpoint; train-validation MSE gap <0.005 confirmed no overfitting.

Asteroid Hazard Classification | *XGBoost, Scikit-learn, SMOTENC*

BTech Major Project

- Built a classifier flagging potentially hazardous Near-Earth Objects on a NASA JPL dataset of **958,524 asteroids** (40 features); **XGBoost** with robust scaling and **SMOTENC** to handle severe class imbalance reached $F1 \approx 0.94$, outperforming Random Forest, Extra Trees and MLP baselines.

Real-Time Facial Expression Recognition | *TensorFlow, OpenCV, CNN*

Personal Project

- Trained a CNN on 35,000+ images across 7 emotion classes (**90% validation accuracy**) and optimised inference to **<80 ms/frame** (25 fps) via model pruning and batch normalisation.

EXPERIENCE**Python Developer Intern**

Jan 2024 – Jun 2024

*Suntist Labs**Delhi, India*

- Engineered a real-time **YOLOv3 + OpenCV** object-detection pipeline, cutting false positives **12%** through iterative hyperparameter tuning on custom labelled datasets.
- Refactored Python preprocessing (Pandas, NumPy) for **30%** higher throughput; unit-tested and deployed 3 production modules in agile sprints, with team-wide deployment documentation.

Data Analytics Trainee

Jun 2023 – Dec 2023

*MedTourEasy**Remote*

- Performed EDA and SQL analysis on **50,000+** patient records and built **Tableau** dashboards used by senior management for planning decisions.
- Streamlined MySQL **ETL** workflows with automated validation checks, substantially reducing manual-entry errors and improving data quality.

CERTIFICATIONS**CS50: Introduction to Computer Science** (Harvard/edX) | **Machine Learning & Automation** (MIT)

AWARDS & ACHIEVEMENTS**2nd Place, Tinkerfest IoT Challenge** – home-automation prototype (embedded sensors + Python control interface) | **3rd Place, N-Crypt N-Code Coding Contest** – algorithmic problem-solving vs 200+ participants